Service Classes
SupabaseClient
+ signUp(email: String, password: String)
+ signIn(email: String, password: String)
+ signOut()
+ query(table: String, filters)
+ subscribe(table: String, callback)
ProfileService
+ getProfile(profileId: UUID) Profile
+ switchRole(profileId: UUID)
+ updateDetails(profileId: UUID, fullName: String)
RouteService
+ distanceM: Int
+ durationS: Int
+ polyline: String
RideService
+ postRide(driverId: UUID, journey, seats: Int) Ride
+ cancelRide(rideId: UUID)
+ completeRide(rideId: UUID)
+ searchRides(filters) List~Ride~
+ getRideHistory(userId: UUID) List~Ride~
MatchingService
- maxPickupDistanceM: Float
- maxDetourM: Float
- timeWindowMin: Int
+ findMatches(request: RideRequest) List~MatchResult~
- isCompatible(ride: Ride, request: RideRequest) Boolean
- calcPickupDistance(ride: Ride, request: RideRequest) Float
- calcDetour(ride: Ride, request: RideRequest) Float
- scoreMatch(ride: Ride, request: RideRequest) Float
RequestService
+ submitRequest(passengerId: UUID, journey) RideRequest
+ chooseRide(requestId: UUID, rideId: UUID)
+ acceptRequest(requestId: UUID)
+ declineRequest(requestId: UUID)
+ cancelRequest(requestId: UUID)
+ getRequestsForRide(rideId: UUID) List~RideRequest~
+ getRequestHistory(userId: UUID) List~RideRequest~
Entity Classes
Profile
+ id: UUID
+ fullName: String
+ role: String
+ createdAt: DateTime
+ switchRole()
+ updateDetails(fullName: String)
1
posts as driver
1
posts as passenger
0..*
0..*
Ride
+ id: UUID
+ driverId: UUID
+ originLat: Float
+ originLng: Float
+ destLat: Float
+ destLng: Float
+ seatsAvailable: Int
+ departureTime: DateTime
+ status: String
+ routePolyline: String
+ routeDistanceM: Int
+ estDurationS: Int
+ hasFreeSeat() Boolean
+ takeSeats(n: Int, passID: UUID)
+ releaseSeats(n: Int)
+ cancel()
+ complete()
1
n..*
recives
RideRequest
+ id: UUID
+ rideId: UUID
+ passengerId: UUID
+ status: String
+ createdAt: DateTime
+ seatsNeeded: Int
+ originLat: Float
+ originLng: Float
+ destLat: Float
+ destLng: Float
+ requestedTime: DateTime
+ chooseRide(rideId: UUID)
+ accept()
+ decline()
+ cancel()
+ complete()
1
0..*
appears in
1
0..*
produces
MatchResult
+ id: UUID
+ requestId: UUID
+ rideId: UUID
+ pickupDistanceM: Float
+ detourM: Float
+ timeDiffMin: Int
+ score: Float
+ createdAt: DateTime